

Assignment

of different properties of water and contributes to the maintenance of life.

Specific property
Good solvent

Contribution to the maintenance of life
allowing many substances to dissolve in it and provides a medium for biochemical reactions inside cells.

Respiratory medium.

Water acts as a respiratory medium. Oxygen dissolves in water, allowing aquatic organisms to obtain oxygen for respiration.

High specific heat capacity.

Another key property of water is its high specific heat capacity. This helps to regulate body temperature preventing sudden temperature changes despite environmental fluctuations.

Transport medium.

It forms the main component of blood, carries nutrients, vitamins and hormones throughout the body. In plants, cohesive and adhesive forces help water move upward to different parts of the plant.

Living environment

Since ice is less dense than water, it floats on the surface, keeping the water below in liquid form. This allows aquatic organisms to survive in cold conditions.

Based on the article, summarise the importance of mineral in maintaining human life using the table below.

Element	Functions	Deficiency symptoms
Inorganic nutrients	Minerals are inorganic nutrients in different amounts, some are needed in large quantities (macro elements) and others in small amounts (trace elements). They are essential for normal growth and metabolism. When minerals are deficient, specific symptoms appear.	
Potassium	maintains ionic balance in cells and supports heart and muscle activity. helps to transmit nerve impulses.	causes muscle weakness and physiological disorders.
Sodium	activates enzymes, forms part of digestive juices and maintains osmotic pressure in cells, assists nerve impulse transmission.	lead to respiratory disorders, cramps, nausea and diarrhea.
magnesium	contributes to bones and teeth and regulates nerve activity skeletal muscles. It supports to metabolic process.	A lack of magnesium may cause high heartbeat and nerve irritability.
Calcium	supports blood clotting, nerve function, milk production and absorption of vitamin B.	results in rickets, weak bones and growth disorders.
Phosphorous.	forms parts of nucleic acid and helps in carbohydrate and fat metabolism.	causes bones to become weak and fragile.
Iron	required for haemoglobin synthesis and oxygen storage in muscles.	leads to anaemia, sleeping and reduced phyceology development.

Iodine	Needed for human thyroid	Deficiency affects intelligence, development, causing lethargy and may limit body height.
--------	--------------------------	---

Minerals maintaining Plant life

Element	Functions	Deficiency symptoms
Nitrogen	Forms part of amino acids, proteins, nucleic acids and chlorophyll	
Phosphorous	Important for root developing	Leads to poor root growth and red or purple patches on leaves
Potassium	Controls the opening and closing of stomata	causes chlorosis and yellow or brown patches on leaves
Sulphur	Forms a part of amino acids and proteins	chlorosis appears in veins and between veins
Iron	Needed for chlorophyll and respiratory enzyme synthesis	causes chlorosis in young leaves
Calcium	Strengthens the cell wall, maintains plasma membrane structure and supports enzyme activity	leads to death of tissues and leaf tips
Zinc	Enzyme activity and chlorophyll synthesis	Deficiency results in dead tissues and thick leaves

Importance of vitamins in the human body

vitamin	uses of the vitamin	Deficiency / symptoms
vitamin A	supports vision and maintain healthy skin.	Night Blindness, Bitot patches, dry skin and respiratory tract disease.
vitamin B	Maintains nerve function, healthy skin, bone marrow formation, red blood cell maturation and antibody production	May cause beri beri, anaemia and reduced immunity.
vitamin D	Controls calcium and phosphorus absorption and supports bone health	May result in tooth decay and osteoporosis.
vitamin C	skin healthy, enamel formation and collagen synthesis.	Weak gums, internal bleeding and scurvy.
vitamin E	supports tissue growth and cell protection.	Cause premature birth, and breakdown of red blood cells.
vitamin K	Required for blood clotting	delays clot formation